Compositional Vector Semantics

Over Different Ground Fields

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Serendipity

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http://code.google.com/p/semanticvectors/

(Just search the web for Semantic Vectors)

Ground Fields and Communities

- Real numbers
- Binary numbers
- Complex numbers

- Statistics / Machine Learning
- Logic / Computing
- Physics

How are they *really* different?

What works for semantics?

“Traditional” Semantic Vector Models

	Doc1	Doc2	Doc3	...
Term1	1	2	3	
Term2	2	0	1	
Term3	1	1	0	
...				

SMARTTFIDFLSAHALGVSMWSDPLSANNMRLDARRI ...

High dimensions implies Sparseness

Compositional Vector Models

- Put together terms into sentences / documents
  - Called “Frege's Principle” but much more Boole
- Lots of available mathematical tools …
  - Linear combination
  - Tensor / Exterior / Grassmannian products
  - Matrices, Projectors, Convolution, Permutations, ...

Motivating Applications

- "Blind Venetian" ≠ "Venetian Blind"
  - (Skip a few decades of disappointment in IR …)
- Negation and Disjunction
- Entailment, WSD, Semantic Parsing, Dependency Modelling, Symbolic Reasoning, Hypothesis Generation, Spell Checking ...

Vector Symbolic Architecture (VSA)

Ross Gayler, Tony Plate, Pentti Kanerva ...

- Vector space V
- Operators x, ●, +, ⊗, ⊘, *generate*(term)
- We want:

$$(a + b) \bullet b \approx 1$$

$$(a \otimes b) \bullet a \approx 0$$

$$(a \oslash (a \otimes b)) \bullet b \approx 1$$

- “Nearly” an axiomatic structure

“Predication Semantic Indexing”

- Inference over Ontologies / Triple Stores
- Corpus example:
  - *usa* CURRENCY *dollar*
  - *mexico* CURRENCY *peso*
- “Train” / “Index” by applying:
  - S(*dollar*) += E(*usa*) ⊗ E(CURRENCY)
  - S(*peso*) += E(*mexico*) ⊗ E(CURRENCY)
  - Etc., modulo discussion of inverse predicates
- Then query by reversing the process …
  - “What's the dollar of Mexico?”

Orthographic Encoding

- Pick a line (e.g., between orthogonal endpoints)
- Subtend points along the line, according to position in the word.
- Create an elemental vector for each character in the alphabet.
- "Encode" the word "bar" as something like:

$$b \otimes 0 + a \otimes 1/2 + r \otimes 1$$

- And then "bar" and "beer" are similar ...

Real Vectors

- Addition and overlap measure are standard vector addition and cosine similarity
- Binding … can choose between:
  - Permutation then addition
    - Fast, huge number of possible permutations, exact inverse
    - Fine for binding one pair – but with several, we lost the constituents! (Shades of entanglement.)
  - Circular convolution
    - Projection of the tensor product
    - O(n log n) with Fast Fourier Transform
    - Commutative
    - Inexact inverse, but it's fine (remember sparseness!)
    - Holographic Reduced Representation, BEAGLE model

Complex Vectors

- Binding with “convolution” is easy …
  - Add the phase angles (Tony Plate, Lance de Vine)
  - O(n) complexity (obviously!)
- So … should addition and similarity measure depend on phase angles too?
- Leads to “flat” and “circular” versions
  - (In this sense, complex numbers and Aristotle's *Physics* are all about the same thing!)

Binary Vectors

- Scaling by a real number isn't normally defined … but we need it!
- Related problem: many equal paths between two points
- Addition with tie-breaking leads to a great deal of tie-breaking with pairs of vectors!
  - Bonus - “orthogonalization”
- Store weights with a “voting record” and then “quantize” as part of running “normalize”
- Bitwise XOR is a natural bundling
  - It's commutative and self-inverse
  - Permuting first can change both of these

Conclusions

- VSAs are (one of several) really interesting models for compositional semantics
- Any number of ways of learning / generating / layering representations
- Groping towards an axiomatization
  - What are the exact rules of recovery, uniqueness, can we formalize “close enough” …
  - “It's a good VSA if dimension > n” sounds terrible to mathematicians … but we're used to it for n = 1
- Always tradeoffs between complexity and fidelity … “reality”, convenience, reuse ...

Main Conclusion!

Don't only use the “real” numbers …

at least, think about it ...

Choice of ground field is really interesting!

Thank you!